The Pythagorean theorem describes right-angled triangles.
In a right-angled triangle the longest side is called the hypotenuse.
The theorem states that a^2 + b^2 = c^2, where c is the hypotenuse.
The square of the hypotenuse equals the sum of the squares of the other two sides.
For example, a triangle with sides 3 and 4 has a hypotenuse of length 5.
This is because 3^2 + 4^2 = 9 + 16 = 25, and the square root of 25 is 5.
Another example is the triangle with sides 5, 12 and 13.
Builders use the theorem to check that the corners of walls are square.
The theorem is named after the Greek mathematician Pythagoras.
It only works for right-angled triangles, not for other triangles.
To find a shorter side, rearrange the formula: a^2 = c^2 - b^2.
For a hypotenuse of 10 and a side of 6, the other side is 8.
The distance between two points on a graph also uses the Pythagorean theorem.
Sets of whole numbers like 3, 4, 5 are called Pythagorean triples.
Students should always label the hypotenuse before using the formula.
Remember to take the square root at the end to get the length of the side.